

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Semester V Examination, 2016-17

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**Paper No. BSC-09A: Statistical Inference, Sampling and Design of Experiments**

Time: 3 Hours

Maximum Marks: 70

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*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

BS/Sem V/183 (Old Course) BSC - 09A : Statistics IT : Statistical Inference, Sampling and Design of Exp. te ae 2. Discuss the t - test for testing the significance of (i) population mean (ii) difference between two population means and (iii) population correlation coefficient. 3. Explain y - test for goodness of fit and for testing the independence of attributes Give examples. a) Describe briefly the advantages of a sample survey in preference to a complete enumeration survey. b) Discuss briefly the basic principles of sampling survey. 4. What is stratified random sampling? Write its advantages and limitations. Discuss any two methods of allocations. a) Define systematic sampling. Describe circular systematic sampling. b) Discuss cluster sampling with its advantages and limitations. 5. Discuss basic principles of design of experiments. Explain CRD and carry out its complete analysis of variance. 6. Write down the layout of RBD and carry out its ANOVA. 7. Write notes on any two of the following : a) Statistical hypotheses, b) F-test, c) Simple random sampling, d) ANOVA. He  
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